^{31}P(^{6}Li,d) 1979Es05

$^{31}P+\alpha \to ^{35}Cl$ from $J^{\pi}=1/2^{+}$ ^{31}P ground state.

1979Es05,1978Es01: a 36-MeV 6Li3^{+} beam at 150-400 nA was produced from the MP tandem accelerator at the University of Rochester and bombarded a 100−μg/cm^{2}cadmium phosphide Cd_3P_2 target. Outgoing deuterons were momentum analyzed by an Enge split-pole magnetic spectrograph and detected using Kodak NTB-50 nuclear emulsion plates with FWHM=50 keV. Measured $\sigma(E_d,\theta)$. Deduced levels, J, π, L-transfers, spectroscopic factors from finite-range, LOLA-DWBA analysis of the measured $\sigma(\theta)$.

Other: 1982Ch10 (finite-range DWBA calculations).

^{35}Cl Levels

Spectroscopic factor $g*S=\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}$, where $g=(2J_f+1)/(2J_i+1)$.

E(level)†	J$^{\pi}$‡	L#	S#
0	3/2^{+}	2	1.00
1220	1/2^{+}	0	1.49
1760	5/2^{+}	2	0.24
3160	7/2^{-}	3	0.52

† From 1979Es05.
‡ As given in 1979Es05 for extracting spectroscopic factors.
# From DWBA analysis of the measured $\sigma(\theta)$ in 1979Es05.